

ORIGINAL ARTICLE

Long-Term Neurological Outcome of Out-of-Hospital Cardiac Arrest Survivors and Associated Factors: Results of the DESAC Study, a French Multicentric Prospective Cohort

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BACKGROUND: The aim of the DESAC study (Devenir des Survivants d'Arrets Cardiaques) was to describe the survival rate and neurological status of out-of-hospital (out-of-hospital cardiac arrest) survivors, and to assess the influence of prehospital and intrahospital factors.

METHODS: Between July 2015 and October 2018, consecutive nontraumatic out-of-hospital cardiac arrest patients aged over 18 years discharged alive from 33 intensive care units in Paris and the suburbs (France) were included. Their neurological status was assessed every 6 months using the Cerebral Performance Category (CPC) scale. Prognostic factors associated with a favorable neurological outcome at 24 months (CPC category of 1 or 2) were assessed by covariate-adjusted logistic regression. Trajectories of CPC level over time were derived using a linear mixed model in survivors with at least 3 repeated assessments.

RESULTS: Out of 593 survivors, 525 were followed up for 24 months (mean age, 58.1±15.3 years, 79% males), and 449 of 525 (86%) had a favorable neurological outcome. Initial shockable rhythm (adjusted odds ratio, 3.33 [95% CI, 1.57–7.07]) and previous practice of sport (adjusted odds ratio, 3.13 [95% CI, 1.21–8.12]) were associated with favorable outcome, whereas factors with reduced odds ratio were older age (adjusted odds ratio per 10 years, 0.53 [95% CI, 0.41–0.69]) and prior percutaneous coronary intervention (adjusted odds ratio, 0.38 [95% CI, 0.17–0.85]). Therapeutic hypothermia or acute coronary interventions were not related to prognosis ($P>0.05$). Overall, among the 514 patients with at least 3 repeated assessments, 5 distinct CPC trajectories (2 reflecting early death were combined) were derived: 64% remained in CPC1, 13.4% in CPC2, 11% improved from CPC2 to CPC1, and 11.6% died either early or late (>3 months) during follow-up.

CONCLUSIONS: This multicentric prospective study indicates that 86% of out-of-hospital cardiac arrest survivors had a favorable neurological outcome at 24 months. Pre-event and intraevent factors (including previous practice of sport and shockable rhythm), but not in-hospital management (therapeutic hypothermia or acute coronary interventions), were associated with a favorable outcome.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT02494414.

Key Words: angiography ■ hypothermia, induced ■ percutaneous coronary intervention ■ out-of-hospital cardiac arrest ■ survivors

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WHAT IS KNOWN

- Despite improvements in recent decades, patients who have experienced out-of-hospital cardiac arrest generally have a poor neurological prognosis, resulting in high mortality rates in the initial phase.
- Only a few patients survive and manage to resume a normal life, while many are left with lasting disabilities. It is difficult to identify and predict specific trajectories of neurological health among these survivors.

WHAT THE STUDY ADDS

- This study including 525 survivors recruited in 33 intensive care units and followed for at least 24 months shows that most survivors recover and maintain favorable neurological conditions. This favorable outcome was associated with younger age, an initial shockable rhythm, and prior participation in sports, but not in hospital management.
- Furthermore, the study identifies distinct patient trajectories of Cerebral Performance Categories during follow-up and the characteristics associated with them. This information is highly relevant to health professionals involved in cardiac arrest management, to policymakers, patients, and their close relatives.

Nonstandard Abbreviations and Acronyms

CA	cardiac arrest
CPC	Cerebral Performance Category
DESAC	Devenir des Survivants d'Arrets Cardiaques
ICU	intensive care units
OHCA	out-of-hospital cardiac arrest
OR	odds ratio
PCI	percutaneous coronary intervention
TTM	targeted temperature management

Overall survival after out-of-hospital cardiac arrest (OHCA) has improved significantly over the past decades.¹ This mainly reflects the improved effectiveness of the chain of survival, and in particular the activation of first responders and the early implementation of cardiopulmonary resuscitation and defibrillation. Advances in post-cardiac arrest (CA) care have also contributed to these results.² However, it should be noted that most clinical studies have focused on a very short-term period (1–6 months) and have often used crude end points such as the rate of return of spontaneous circulatory activity or overall survival.^{3,4} It is not definitively established whether the improvement in these indicators is accompanied by

an improvement in neurological prognosis in the short and long term.¹

Looking more specifically at the available data in CA survivors, it is generally established that their overall survival is satisfactory at 1 year, but little is known about modifiable or nonmodifiable prognostic factors.^{5,6} However, early identification of survivors who will be at greatest risk of developing significant long-term sequelae could guide rehabilitation and treatment strategies. In addition, early identification of factors that influence long-term prognosis could help clinicians in their care of these survivors and lead to active prevention.⁷ Furthermore, available studies have not considered how prognosis evolves over time.⁸ It would be useful to better understand the possibilities of evolution and recovery over time, and to know whether certain initial care parameters are associated with subsequent trajectories. A better understanding of the effect of care strategies on long-term outcomes could help target adequate allocation of resources to patients and understand the future of patients and their families.

To address these issues, we set up a comprehensive and regional cohort of CA survivors, the DESAC cohort (Devenir des Survivants d'Arrets Cardiaques). Our objectives were to describe the neurological status at 24 months, to identify the different trajectories of evolution of the neurological status over time, and to evaluate the influence of pre- and intrahospital factors.

METHODS

The statistical code and data that support the findings of this study are available from the corresponding author on reasonable request. The study was approved by the institutional review committee, Comité de Protection des Personnes Paris Ile de France IV, according to the article L.1123-6 from the French Public Health code, no. 2012-A01515-38 and the collection of individual data was simultaneously declared to the Commission Nationale de l'Informatique et Libertés according to the articles L1121-1 et seq from the French Public Health code (no. 914505).

The DESAC study was a multicentric, prospective, population-based study in which patients who survived an OHCA and were discharged alive from the hospital were subsequently followed using regular phone interviews. The detailed design of the study was recently published.⁹

Population

Between July 2015 and October 2018, consecutive patients over 18 years old, survivors, that is, discharged alive from 33 intensive care units (ICU) in Paris and suburbs area (France) after a nontraumatic OHCA were included after providing written consent. The calculation of the number of subjects to be included was performed to assess the long-term effect of therapeutic hypothermia, which at the time of the study's design was one of the therapeutic interventions likely to improve the neurological outcome of these survivors. Assuming a statistical

power of 80%, an alpha risk of 5%, and a 2-sided test, we calculated that 292 subjects per group (594 patients in total) would be required to observe at 24 months a 10% difference in favor of hypothermia in the recovery of a favorable neurological status, as assessed by the Cerebral Performance Category (CPC) scale.¹⁰ Exclusion criteria were impossibility of communication after discharge from the hospital (related to language or geographic distance), refusal of follow-up by patients or their relatives, and the usual place of residence outside the Paris region.

Follow-Up

After ICU's discharge (follow-up start, time 0), patients were followed up by repeated telephone interviews over time. The first interview took place at the third month after OHCA, then at the sixth month and every 6 months thereafter. A research assistant, trained in telephone interviews, called the patients to submit the different questionnaires. Senior investigators from the Sudden Death Expertise Centre supervised the training and qualitative completion of the data through regular follow-up. Each participant was asked to give their consent at the time of their inclusion in the study. If the patient's condition did not allow them to understand and give consent at the time of inclusion, a relative could give consent on their behalf. Once initial consent had been obtained from a relative, consent from the patient was systematically sought during follow-up, depending on their neurological improvement. In severely disabled patients, telephone interviews were conducted with their representatives. To minimize the risk of cohort attrition, up to 6 attempts to reach each patient were made on different days and times. For patients lost to follow-up and unable to be contacted, their vital status was periodically assessed (at the time of each initially scheduled interview) by checking the public death register of the commune of birth. Follow-up was censored at the time of death, loss to follow-up, patient withdrawal, or study end, whichever occurred first.

End Points

The main end point was neurological status at 24 months, assessed by the Pittsburgh CPC scale (Table S1). Briefly, this scale allows to classify patients according to 5 levels, from level 1 for normal or minimal disability (patient able to care for himself, discharged directly to home) to level 4 for severe disability (unconscious, discharged to a long-term nursing facility) or even 5 for death or brain death. In this study, a favorable neurological evolution was defined as CPC levels 1 and 2. The definition of early death corresponds to a death occurring within the first 3 months after CA.

Data Collection

Data were collected according to the Utstein style¹¹ including patient characteristics (age, cardiovascular risk factors, sports practice), event characteristics (location of CA, presence of a witness, initial heart rate, use of a defibrillator, duration of resuscitation), initial treatments (defibrillation, epinephrine or antiarrhythmic treatments), appearance of the initial ECG and presumed cause of arrest, as well as hospital management modalities (including early coronary angiography and targeted temperature management [TTM]). Importantly, sport

practice corresponds to usual practice before OHCA but not circumstantial sport practice at the time of OHCA. Participating centers were encouraged to apply the French and European guidelines in place at the time of the study⁴ regarding TTM and coronary angiography. These guidelines encouraged TTM between 32°C and 36°C, as well as early coronary angiography whenever a coronary cause was suspected.

Statistical Analysis

We used descriptive statistics to assess characteristics according to the primary outcome (CPC1–2 versus CPC3–5 at 24 months). They were summarized as proportions for categorical variables and as medians (interquartile range) and means (SD) for continuous variables. Comparisons were made using the χ^2 test for categorical variables and Student or Wilcoxon tests for continuous variables.

We performed a multivariate logistic regression analysis of a favorable outcome at 24 months, including factors deemed significant ($P < 0.15$) after univariate analysis. We then performed a backward stepwise logistic regression analysis using a threshold of $P < 0.05$ to stay in the model, adjusting for the variables of interest, sex, percutaneous coronary intervention (PCI), and TTM. PCI and TTM were selected to meet our initial hypothesis on their potential influence on the outcome. The analysis was stratified by ICU to account for heterogeneity in terms of volume of activity, patients' characteristics, and possible differences in clinical management. We used a complete case analysis.

We described the trajectories of the CPC score over time using the repeated assessments over 24 months of follow-up. We used a group-based trajectory modelling implemented by SAS Proc Traj, using the censored normal model. We included participants who had at least 3 CPC measurements, including those at resuscitation discharge and 1 year, and one measurement at 3 or 6 months. The optimal number of trajectories was based on statistical criteria, including the Bayesian Information Criterion, and clinical parameters, including the number of patients per trajectory (at least 10/trajectory). The derivation of the CPC trajectories is described in detail in the [Supplemental Material](#). The baseline characteristics and circumstances surrounding the CA (Utstein factors) that remained associated with CPC trajectories were evaluated in multinomial logistic regression analysis, with the stable CPC1 trajectory as the reference, and including a random effect on ICU.

In sensitivity analysis, we repeated the analysis of the factors associated with CPC1 to 2 at 24 months using multiple imputations to account for missing covariates. Data were considered to be missing at random. The imputation procedure used covariates thought to be associated with the missingness mechanism and all the covariates included in the logistic regression model. Multivariate imputation by the fully conditional specification method was carried out using 10 imputations (20 burn-in iterations). The same approach was used when further imputing missing CPC at 24 months.

Finally, the multivariable analysis of risk factors for CPC1 to 2 at 24 months was stratified by TTM status, and formal interactions between each risk factor and TTM status were evaluated using a P value of $< 10\%$ as a statistical threshold.

The validity of the multivariate multinomial logistic regression was checked and verified (P value of the likelihood ratio

test, $P<0.0001$; Hosmer-Lemeshow test, $P=0.17$, variance inflation factor ranging from 1.16 to 1.42, indicating low correlation between the predictors).

Unless otherwise stated, all tests were 2-sided with a P value considered significant if <0.05 and were performed using SAS software 9.4 (Statistical Analysis System, Cary, NC).

RESULTS

Between July 2015 and October 2018, 593 patients who were discharged alive from the 33 participating ICUs after a nontraumatic OHCA were enrolled. Among them, 525 (89%) underwent a neurological status evaluation at 24 months and were eligible for the main analysis (Figure 1). The characteristics of those with and those without available neurological status at 24 months are compared in Table S2. Of those with available follow-up, 449 (86%) had a favorable neurological outcome as defined by a CPC1 to CPC2 rating (399 in CPC1 and 50 in CPC2), and 76 (14%) had an adverse outcome (CPC3–5) at 24 months. As shown in Table 1, at 24 months, patients with CPC1 to CPC2 were younger, more likely to have been arrested in public, to have shockable rhythms, and to have cardiac etiology. In addition, they were less likely to have comorbidities such as hypertension, diabetes, or previous acute coronary

syndrome. There was no difference in TTM management between groups ($P=0.38$), whereas PCI was performed more frequently in CPC1 to CPC2 patients at 24 months ($P=0.0007$). After multivariable analysis (Table 2), initial shockable rhythm (odds ratio [OR], 3.33 [95% CI, 1.57–7.07]) and practicing sport before the CA (OR, 3.13 [95% CI, 1.21–8.12]) were positively associated with achieving a CPC1 to CPC2 at 24 months. On the opposite, aging (per 10 years increment, OR, 0.53 [95% CI, 0.41–0.69]) and a history of PCI (before the occurrence of CA) were inversely associated with CPC1 to CPC2 outcome (OR, 0.38 [95% CI, 0.17–0.85]). Of note, in-hospital management, including PCI and TTM during the postresuscitation period, was not associated with CPC1 to CPC2 at 24 months. Patients exposed to TTM did not differ from those unexposed except that they were more often current smokers, on shockable rhythm, diagnosed with a presumed cardiac etiology, but had higher no-flow and low-flow duration (Table S3).

The number of patients with available CPC over 24 months is reported in Table S3, whereas the Sankey diagram showing the change in CPC category over 24 months is reported in Figure 2. Of the 593 patients initially recruited, CPC level could be assessed in 556 at hospital discharge, in 568 at 3 months, 471 at 6 months, 551 at 12 months, 518 at 18 months, and 525 at 24

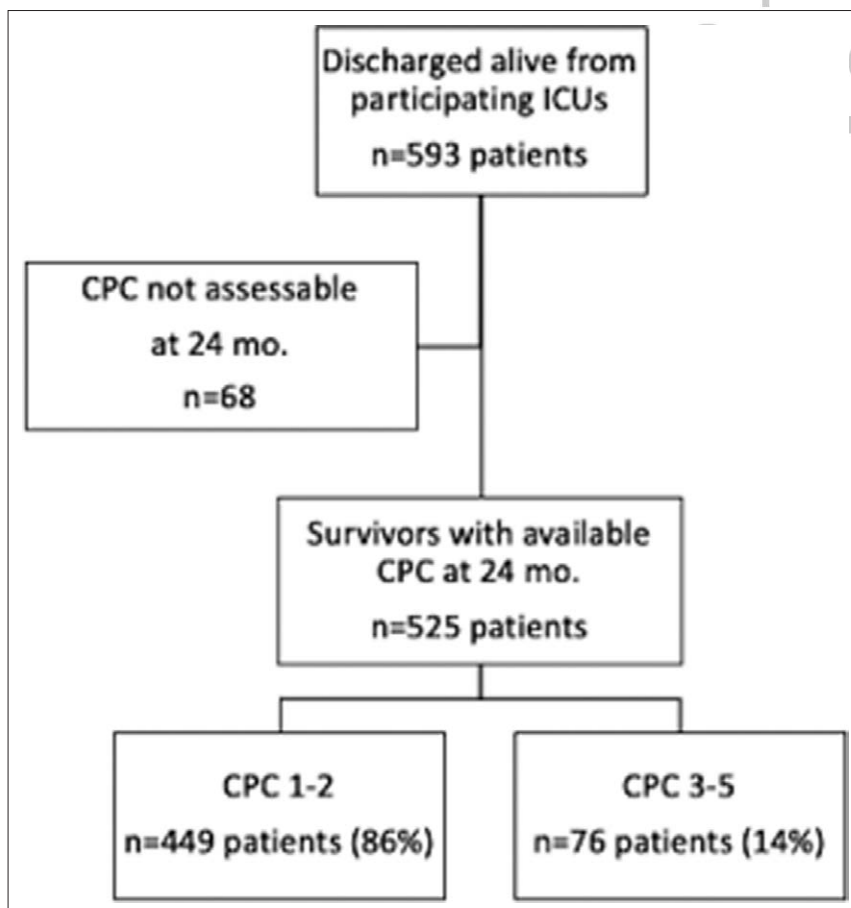


Figure 1. Study flow chart.
CPC indicates Cerebral Performance Category; and ICU, intensive care unit.

Table 1. Baseline Characteristics According to CPC Level at 24 Months

	CPC at 24 mo (N=525)			P value
	Overall	CPC1–2	CPC3–5	
	N=525	n=449	n=76	
Age, y	58.11 (15.3)	56.43 (14.9)	67.99 (14.4)	<0.0001
Gender (male)	416 (79.2)	356 (79.3)	60 (78.9)	0.95
SP categories				0.01
Farmer	3 (0.6)	2 (0.5)	1 (1.4)	
Craftsman	43 (8.8)	39 (9.3)	4 (5.6)	
Executive employee	89 (18.1)	83 (19.8)	6 (8.4)	
Intermediary occupation	28 (5.7)	25 (5.9)	3 (4.2)	
Employee	84 (17.1)	77 (18.3)	7 (9.9)	
Worker	29 (5.9)	25 (5.9)	4 (5.6)	
Unemployed	50 (10.2)	41 (9.8)	9 (12.7)	
Retired	165 (33.6)	128 (30.5)	37 (52.1)	
Married or cohabiting	361 (71.9)	310 (72.3)	51 (69.9)	0.67
Sport practice prior CA	122 (24.9)	115 (27.6)	7 (9.6)	0.001
Smoking				0.32
Nonsmoker	200 (39.7)	172 (39.9)	28 (38.4)	
Current smoker	192 (38.1)	168 (39.0)	24 (32.9)	
Exsmoker	112 (22.2)	91 (21.1)	21 (28.8)	
Hypertension	193 (38.4)	151 (35.1)	42 (57.5)	0.0003
Dyslipidemia	134 (26.7)	115 (26.8)	19 (26.0)	0.89
Diabetes	66 (13.1)	47 (10.9)	19 (26.4)	0.0003
Family history of CVD	93 (18.7)	87 (20.5)	6 (8.3)	0.01
Under chronic medication	302 (60.0)	242 (56.1)	60 (83.3)	<0.0001
PCI prior event	60 (11.5)	42 (9.7)	18 (25.3)	0.0002
PAD	16 (3.2)	12 (2.8)	4 (5.6)	0.27
Rhythmic disorder	79 (15.8)	61 (14.2)	18 (25.3)	0.02
Previous stroke	24 (4.8)	17 (3.9)	7 (9.7)	0.06
CA location				
Public	251 (50.0)	225 (52.3)	26 (36.1)	0.002
Home	206 (41.0)	163 (37.9)	43 (59.7)	
Other	45 (9.0)	42 (9.8)	3 (4.2)	
Witness	478 (95.4)	411 (95.6)	67 (94.4)	0.55
Bystander CPR	394 (84.5)	342 (85.5)	52 (78.8)	0.16
Shockable rhythm	392 (78.7)	353 (82.7)	39 (54.9)	<0.0001
No flow	1.00 (0.00–3.00)	1.00 (0.00–3.00)	0.00 (0.00–3.00)	0.22
Low flow	15.00 (7.00–22.00)	15.00 (7.00–21.00)	13.00 (6.00–23.00)	0.67
Presumed etiology of CA				<0.0001
Cardiac	420 (84.2)	372 (87.1)	48 (66.7)	
Neurological	2 (0.4)	1 (0.2)	1 (1.4)	
Respiratory	45 (9.0)	28 (6.6)	17 (23.6)	
Toxic	4 (0.8)	3 (0.7)	1 (1.4)	
Other	28 (5.6)	23 (5.4)	5 (6.9)	
Epinephrine use	180 (36.4)	145 (34.3)	35 (48.6)	0.02
Plasma creatinine, $\mu\text{mol/L}$	93.00 (77.00–119.00)	92.00 (77.00–113.00)	111.00 (86.00–146.00)	0.0002
ICU TTM	314 (63.3)	273 (64.1)	41 (58.6)	0.38
PCI after CA	236 (48.2)	215 (51.3)	21 (29.6)	0.0007

(Continued)

Table 1. Continued

	CPC at 24 mo (N=525)			P value
	Overall	CPC1–2	CPC3–5	
	N=525	n=449	n=76	
CPC at ICU discharge*				<0.0001
1–2	450 (90.9)	398 (93.9)	52 (73.2)	
3–4–5	45 (9.1)	26 (6.1)	19 (26.8)	
Deceased	90 (17.1)	18 (4.0)	72 (94.7)	

Data are presented as no. of patients (n) and percentages (%)—excluding missing data—or mean (SD) where appropriate; median (q1–q3) is reported for low-flow, no-flow, and creatinine concentrations. P values derived from unadjusted Pearson χ^2 or t test. CA indicates cardiac arrest; CPC, Cerebral Performance Category; CPR, cardio-pulmonary resuscitation; CVD, cardiovascular disease; ICU, intensive care unit; PAD, peripheral artery disease; PCI, percutaneous coronary intervention; SP, socio-professional; and TTM, targeted temperature management.

*Information on CPC at ICU discharge was missing for 30 patients, explaining that the total reached n=495 instead of N=525 patients.

months. Overall, 514 (87%) had at least 3 assessments over 24 months and were eligible for the CPC trajectory analyses. From the 6 identified distinct trajectories of CPC levels, 2 reflecting early death (trajectory 1+6) were grouped to achieve a total of 5 trajectories (Figure 3). These include 64% survivors with stable CPC1, 13.4% with stable CPC2, 11% who improved from CPC2 to CPC1, 9.1% who died early, and 2.5% who died late, respectively. The distribution of the characteristics by CPC trajectories over 24 months is described in Tables S4–S5. Factors associated with CPC trajectories in multivariate analysis are shown in Table 3. In particular, older age (OR, 1.62 [95% CI, 1.25–2.10]), the use of epinephrine during initial resuscitation (OR, 2.91 [95% CI, 1.43–5.94]) and the absence of an initial shockable rhythm (OR, 4.16 [95% CI, 1.82–9.09]) were more likely related to early death trajectory than stable CPC1 trajectory, whereas older age was more likely associated with the late death trajectory (OR, 2.55 [95% CI, 1.40–4.62]) than the stable CPC1 trajectory.

In sensitivity analysis, after multiple imputations, the factors associated with CPC1 to 2 at 24 months, together with the effect size, were consistent with the

results of the main analysis (Tables S6 and S7). In addition, the risk factor analysis for CPC1 to 2 at 24 months by TTM status showed consistent results in each stratum (Table S8). However, statistical significance was not reached for prior sport practice among those unexposed to TTM, and for shockable rhythm and PCI before OHCA in those exposed to TTM, respectively.

Furthermore, the factors associated with CPC1 did not differ from those associated with CPC2. Lastly, CPC trajectories up to 48 months were similar to those up to 24 months, including 69.4% (n=357) in stable CPC1, 15.9% (n=82) in stable CPC2, 11.7% (n=60) reflecting early death trajectory, and 3% (n=15) indicating late death trajectory. Still, and contrary to the main analysis, there was no trajectory reflecting improvement from CPC2 to CPC1 (Figure S1).

DISCUSSION

Analysis of this multicentric cohort of 593 CA survivors recruited in 33 French ICUs revealed 3 main findings. First, most survivors (86%) had a favorable neurological status (CPC1–2) at 24 months. Second, younger age, initial shockable rhythm, and previous sport practice before the event were the factors significantly associated with a favorable neurological status (CPC1–2) at 24 months, whereas no association was seen with in-hospital intervention, including early coronary interventions and temperature management. Third, and for the first time, we derived trajectories of CPC over time and identified 5 distinct patterns, including those who remained neurologically stable as either CPC1 or CPC2, those who improved their CPC score, and those who irreversibly progressed towards early or late death. In particular, two-thirds stayed on CPC1, and 11% improved from CPC2 to CPC1.

Long-term survival (classically after 1 year of follow-up) after CA has already been investigated, showing satisfactory survival rates. A recent systematic review highlighted a survival rate at 1 year of almost 80% based on 41 studies and a survival rate of 68.5% at 3 years

Table 2. Multivariable Analysis of Predictors of CPC1 to CPC2 at 24 Months

	Unadjusted OR (95% CI)	Adjusted OR (95% CI)
Age, per 10 y increase	0.53 (0.42–0.66)	0.53 (0.41–0.69)
Sex, male vs female	0.94 (0.51–1.73)	0.83 (0.38–1.83)
Sport practice prior event	4.08 (1.70–9.81)	3.13 (1.21–8.12)
Shockable rhythm	3.69 (2.08–6.55)	3.33 (1.57–7.07)
PCI prior event	0.31 (0.16–0.60)	0.38 (0.17–0.85)
TTM	1.36 (0.77–2.41)	0.85 (0.43–1.71)
PCI after CA	3.10 (1.74–5.53)	1.60 (0.79–3.28)

ORs and 95% CI derived from backward binary logistic regression analysis adjusted for the variables listed in the table. This complete case analysis was conducted in 468 out of the 525 eligible patients due to missing covariates in 57 patients. CA indicates cardiac arrest; CPC, Cerebral Performance Category; OR, odds ratio; PCI, percutaneous coronary intervention; and TTM, targeted temperature management.

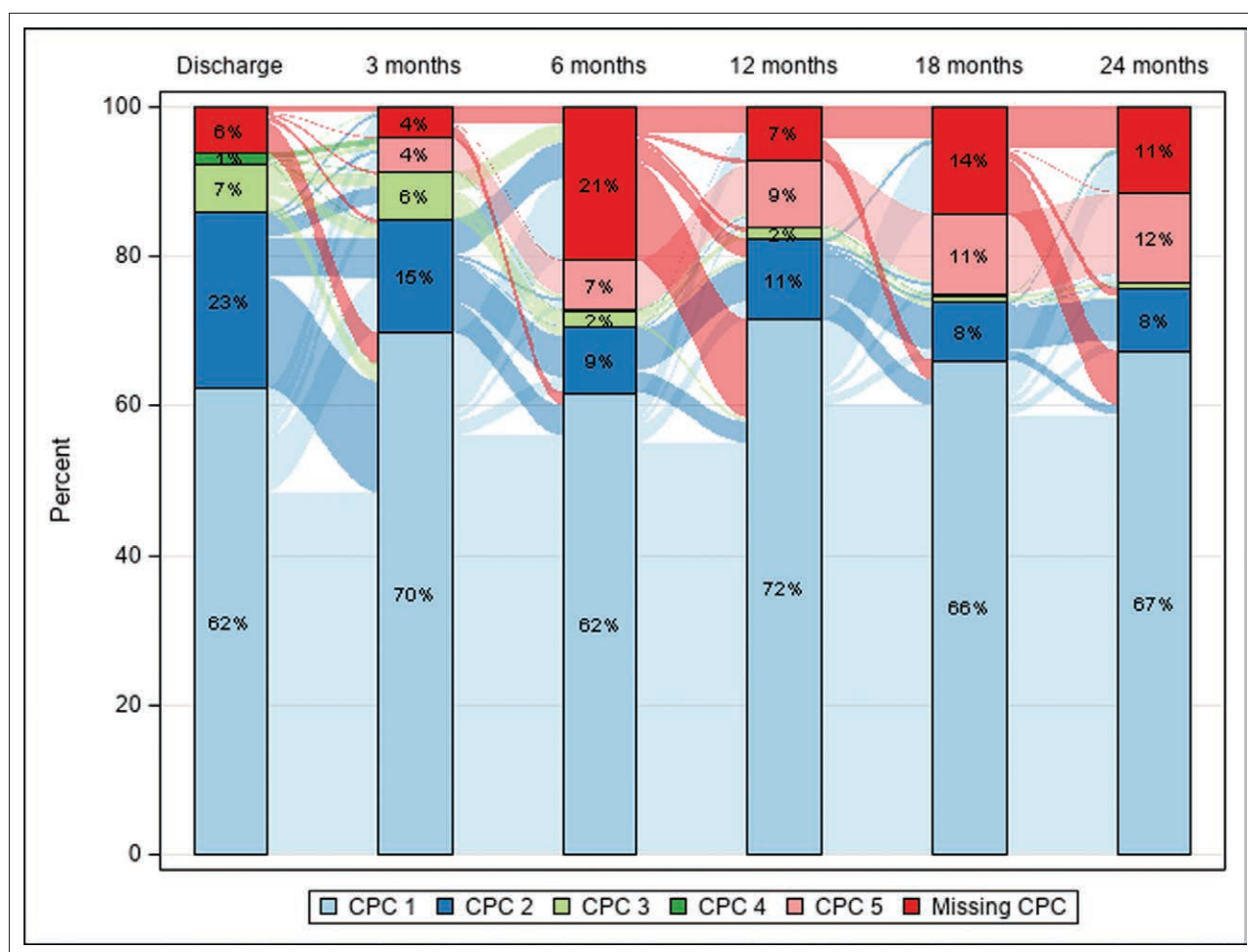


Figure 2. Sankey diagram of Cerebral Performance Category (CPC) change over 24 months of follow-up.

based on 15 studies of nontraumatic CA, although there was a wide heterogeneity of survival across publications.¹² The survival rates of 86% and of 80%, respectively at 24 months and at 3 years in the current study are consistent and even higher compared with prior studies. We provided additional evidence by examining the long-term functional status of OHCA survivors and reported that, respectively, 76% and 10% of OHCA survivors had a CPC1 and CPC2 at 24 months, reflecting good neurological status. The few prior studies on functional outcomes in OHCA survivors (cognitive impairment or social participation) were mostly conducted in selected clinical samples enrolled in randomized controlled trials evaluating the benefit of therapeutic hypothermia, and addressed short-term prognosis, typically 6 months.^{13–16}

We also investigated the determinants of good neurological outcome at 24 months. To the best of our knowledge, this is the first study evaluating the association between sports practice before OHCA and long-term neurological status. Sport practice and regular physical activity have been consistently associated with a lower risk of OHCA in the general population.¹⁷ The current study extends this evidence by suggesting that

sport practice before OHCA is associated with good neurological status in OHCA survivors at 24 months, showing a large effect size (OR, 3.38) that is independent of Utstein factors. Comparatively, in a small Norwegian study of 175 OHCA survivors, a subgroup analysis indicated that the CPC level at hospital discharge was significantly higher among those who exercised 5h or more per week before CA compared with those who exercised less.¹⁸ Furthermore, in addition to younger age, the significant association of shockable rhythm with good neurological status at 24 months in the present study confirms the importance of the chain of survival. This finding extends the results of several prior studies, including a large Canadian cohort investigating the prognostic value of Utstein factors for long-term mortality risk in CA survivors.⁵ Although PCI and TTM have been related to better long-term survival in large cohorts,^{5,19} the evidence is less clear regarding neurological outcomes as reported in the present study and in other studies.^{13,14} Interestingly, after multivariable analyses, epinephrine use was no longer associated with long-term outcome. Although this factor is classically associated with an adverse postresuscitation

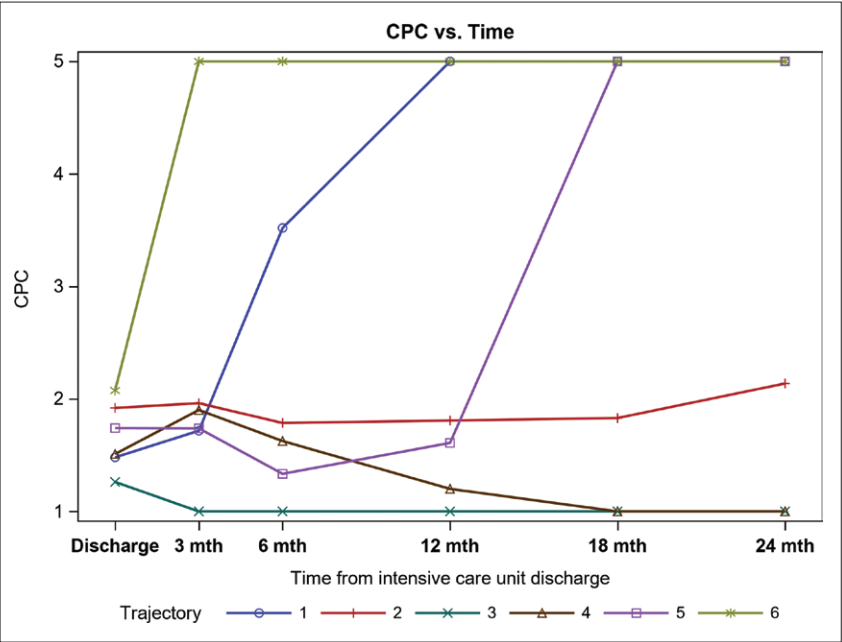


Figure 3. Cerebral Performance Category (CPC) trajectories at repeated assessments over 24 months of follow-up. CPC ranges from 1 to 5, with the lower the better neurological status, see Table S1. Trajectory 1+6: early death; trajectory 2: stable CPC2; trajectory 3: stable CPC1; trajectory 4: improvement from CPC2 to CPC1; and trajectory 5: late death.

effect, its association with long-term outcome is unclear. However, it is noteworthy that information on the dosage of epinephrine was unavailable in the current study. To the best of our knowledge, this is also the first study describing the long-term dynamics (ie, trajectories) of neurological status based on repeated assessments of CPC over 24 months. This analysis indicates that the 86% of OHCA survivors with CPC1 to 2 at 24 months reflect two-thirds with stable CPC1, 13% with stable CPC2, and 11% who improved their neurological status from CPC2 to CPC1. This analysis also permitted the identification of factors associated with the early death trajectory, representing 9% of the OHCA survivors. The few existing studies with repeated assessments had follow-up limited to 6 months, but consistently with our study showed a favorable cognitive progression.^{8,20,21} Taken together, studies like ours describing good long-term neurological status among OHCA survivors

are crucial for the patient, for his family, and for his close relatives. It is also crucial to guide and support resource allocation and effective interventions not only for acute care but also for rehabilitation after hospital discharge. This corroborates the result of a recent retrospective survey among survivors of OHCA in Denmark, showing that the long-term health-related quality of life up to 20 years after their event was consistently high and comparable to that of the general population.²² Furthermore, interventions such as early screening for cognitive and emotional problems, promotion of self-management, and referral to specialized care if necessary after CA have been associated with a lower risk of long-term physical and cognitive impairment.²³ Despite the long-term and repeated assessment of the neurological status of OHCA survivors, we acknowledge some limitations in the current study. First, although the survivors were recruited from 33 ICUs in the Paris area, representing 11% of the French population, the

Table 3. Multivariable Analysis of Predictors of CPC Trajectories Over 24 Months

	Stable CPC1; n=47	Stable CPC2; n=69	CPC2 to CPC1; n=329	Early death; n=57	Late death; n=12
Adjusted OR (95% CI)					
Age, per 10 y increase	1 (Ref)	1.24 (1.02–1.50)	1.17 (0.96–1.44)	1.62 (1.25–2.10)	2.55 (1.40–4.62)
Sex, male vs female	1 (Ref)	0.65 (0.33–1.26)	0.80 (0.39–1.65)	1.13 (0.48–2.70)	1.11 (0.20–6.19)
Bystander CPR	1 (Ref)	0.30 (0.15–0.62)	0.43 (0.19–0.99)	0.50 (0.20–1.25)	NA
Shockable rhythm	1 (Ref)	1.13 (0.49–2.61)	1.29 (0.52–3.16)	0.24 (0.11–0.55)	0.58 (0.12–2.87)
Epinephrine use	1 (Ref)	2.82 (1.60–4.99)	1.16 (0.60–2.24)	2.91 (1.43–5.94)	2.67 (0.65–10.96)
TTM	1 (Ref)	0.99 (0.55–1.80)	0.17 (0.56–2.42)	0.92 (0.45–1.90)	2.99 (0.55–16.17)
PCI after CA	1 (Ref)	1.03 (0.56–1.90)	0.42 (0.21–0.84)	0.70 (0.30–1.62)	0.30 (0.05–1.68)

ORs and 95% CI derived from backward multinomial logistic regression analysis using stable CPC1 as the reference category and adjusted for the variables listed in the Table and including a random effect on the intensive care units. This complete case analysis was conducted in 464 out of the 514 eligible patients due to missing covariates in 50 patients. In this table, trajectories 1 and 6 from Figure 3 have been grouped into a single category labeled early death. CA indicates cardiac arrest; CPC, Cerebral Performance Category; CPR, cardiopulmonary resuscitation; OR, odds ratio; PCI, percutaneous coronary intervention; NA, not applicable; Ref, reference; and TTM, targeted temperature management.

results of the DESAC cohort may not be generalizable, especially to areas with much less medical density. Second, there were 11% of patients who were lost to follow-up for the primary outcome at 24 months. However, they were younger and more often female compared with those with follow-up data, minimizing the risk of positive selection bias. Still, this may have affected the statistical power of our analysis. Comparatively, other studies monitoring long-term outcomes in this area have reported similar or even higher rates of loss to follow-up.²⁴ Third, although being routinely used by the clinical management teams, the CPC scale may not be sensitive enough to detect subtle cognitive impairment. Furthermore, additional important functional outcomes such as quality of life, depression and anxiety, or return to work were not examined.^{25,26} Fourth, the proportion of sports practice before CA was limited and should temper the conclusion regarding the association between this factor and outcome. In addition, we did not have detailed information, such as frequency, intensity, or duration of sports practice. Fifth, telephone-based CPC assessments may reduce accuracy, especially in disabled or cognitively impaired individuals. Proxy interviews can introduce misclassification bias. Sixth, we cannot exclude that non-significant associations may reflect a lack of statistical power because the sample size calculation was based on the expected benefit of therapeutic hypothermia for CPC at 24 months and not for risk factors for favorable CPC at 24 months.

CONCLUSIONS

The DESAC study is one of the largest prospective studies examining the neurological outcome of CA survivors 24 to 48 months after the event and the associated prognostic factors. Such evidence is of considerable importance to stakeholders in CA management, policy makers, the patients, and their close relatives. Additional studies exploring the long-term prognosis of OHCA survivors in different contexts are needed.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S8

Figure S1

STROBE Checklist



APPENDIX

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